Outlier Analysis: HR Employee Attrition

Dataset: 999 employees. Overall attrition rate: 16.7%. Method: 1.5 x IQR rule, run in Power BI (DAX) and in Python (pandas).

1. Summary

- I checked ten numeric columns for outliers using the 1.5 x IQR rule.
- Outliers exist in 7 of the 10 columns, and every one of them is on the high side, with no low-side outliers.
- I then tested whether the outliers are real. They are: no logical rule is broken in any column, and no employee has more years at the company than years since age 18.
- **Decision: I kept every outlier.** They are valid values inside a normal range for this workforce, for example a long-serving employee with 40 years of tenure, or a director on a senior salary. Removing them would distort the picture.
- I built the first version in Power BI, then moved the analysis to Python because the output and the charts were clearer (section 6).

2. Method

For each column the fences are Q1 - 1.5 x IQR and Q3 + 1.5 x IQR, where IQR = Q3 - Q1. A value outside the fences is flagged. A flag means unusual, not wrong. I then asked two questions of every flagged group: is the value possible, and does it change the attrition story?

3. Results by column

Column	Q1	Q3	Upper fence	Outliers	Share	Max	Attrition in outliers
Monthly income	2,865	8,704	17,462.5	64	6.4%	19,999	8%
Years at company	3	9	18.0	76	7.6%	40	11%
Total working years	6	16	31.0	27	2.7%	40	11%
Years since promotion	0	3	7.5	77	7.7%	15	13%
Years in current role	2	7	14.5	14	1.4%	18	14%
Years with manager	2	7	14.5	12	1.2%	17	0%
Companies worked	1	4	8.5	41	4.1%	9	22%
Age	30	43	62.5	0	0.0%	60	n/a
Distance from home	2	14	32.0	0	0.0%	29	n/a
Salary hike %	12	18	27.0	0	0.0%	25	n/a

Attrition among all employees is 16.7%. Low-side outliers: 0 in total.

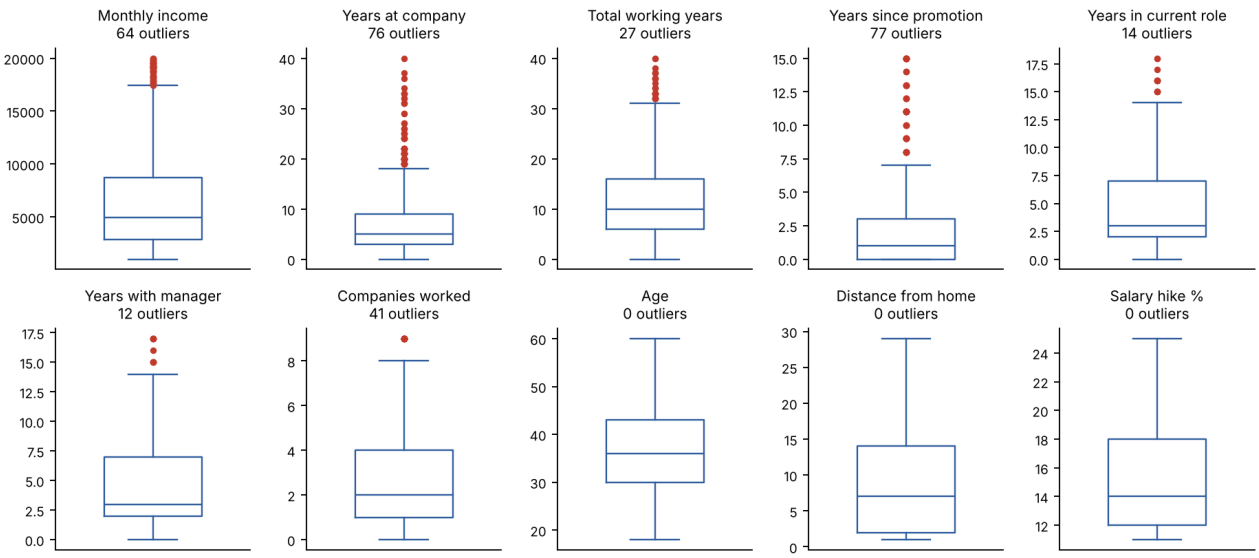


Figure 1. Box plots for the ten columns. Red points are outliers.

4. Are the outliers real?

I tested the data for impossible values. A real error would show up as a broken rule. None did.

Check	Rows failing
Years at company greater than total working years	0
Total working years greater than age minus 15	0
Years in current role greater than years at company	0
Years since promotion greater than years at company	0
Years with manager greater than years at company	0
Income of zero or below	0
Age below 18	0
Missing values (all columns)	0
Duplicate rows	0

Tenure outliers.

- 76 employees have unusually long tenure (above 18 years). The longest is 40 years.
- All of them started working no earlier than age 18 (0 rows break this rule), so the values are possible. Long service is normal in a company with senior staff.
- Their attrition rate is 11%, below the company rate of 16.7%. They do not drive attrition.

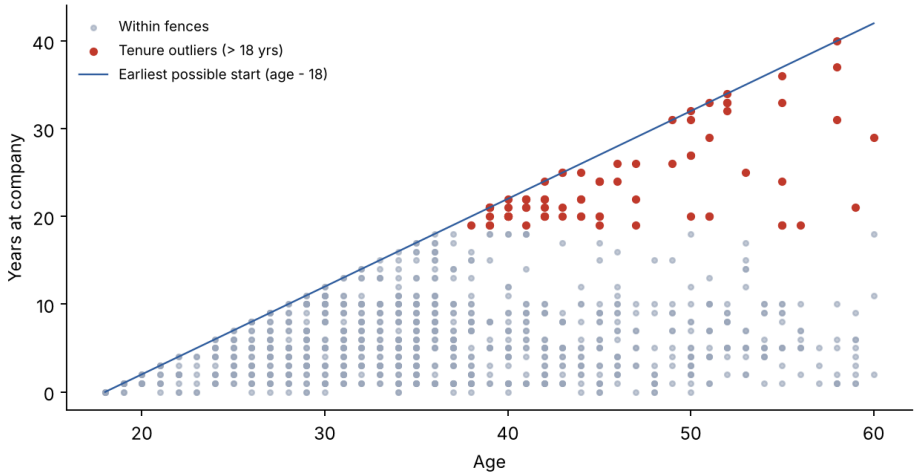


Figure 2. Years at company against age. Every point sits above the line, so tenure never exceeds the possible maximum.

Income outliers.

- 64 employees earn above 17,462 a month (the maximum is 19,999).
- They sit at the top job levels: level 4 (10), level 5 (54). By role: Manager (40), Research Director (24).
- Their attrition rate is 8%. High earners are staying, so the retention problem sits at the low end of pay.

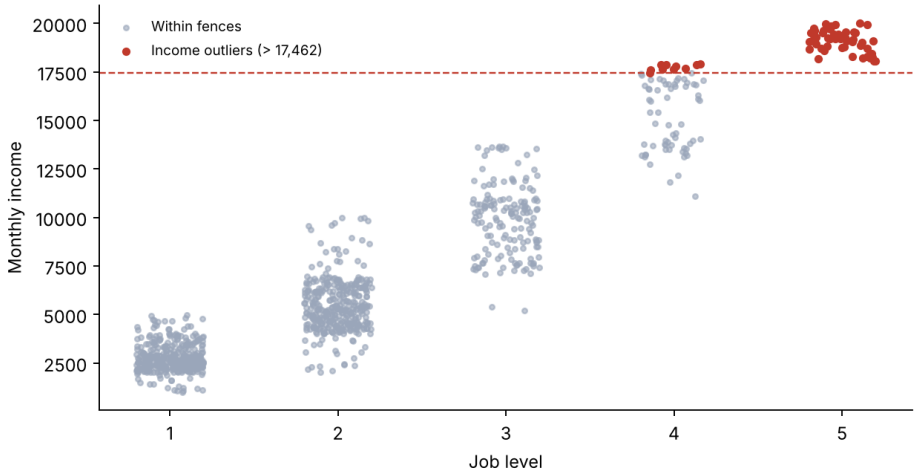


Figure 3. Monthly income by job level. Red points are above the upper fence; they cluster at levels 4 and 5.

5. Decision: keep the outliers

Group	What I found	Action
Tenure and working years	Valid long careers. No rule broken.	Kept
Income	Genuine senior pay at levels 4 and 5.	Kept. Used the median for comparisons.
Years since promotion, years in role, years with manager	Long-serving staff. All consistent with tenure.	Kept
Companies worked	Real job histories. Employees with many past employers leave somewhat more often.	Kept

Why I did not remove or cap them:

- They are not errors. Removing valid people changes the headcount and the attrition rate.
- Deleting high earners would hide the real finding: leavers are low paid, not high paid.

- Capping (winsorising) would be reasonable for a predictive model. This is a descriptive analysis, so I report the data as it is.

How I protected the analysis from them:

- I compared income with the median instead of the mean. The mean is pulled up by the senior salaries.
- I kept the outlier threshold and count live in the dashboard, so any viewer can see them for any filter.
- Effect on the mean: monthly income is 6,617 with outliers and 5,775 without. The median is 4,936, so I used it.

6. Why I moved from Power BI to Python

	Power BI (DAX)	Python (pandas, matplotlib)
IQR calculation	Possible, but each fence needs its own measure and table.	One loop covers every column.
Visuals	Box plots need an add-on visual. Limited control.	Box plots, scatter plots and highlighted outliers with full control.
Plausibility checks	Hard to express across several columns.	Simple row-level rules, with counts.
Output	Good for interactive filtering.	Tables and a printable report, repeatable on new data.

I kept Power BI for the interactive dashboard, where the outlier threshold and count respond to slicers. I used Python for the analysis and the evidence.

7. Conclusion

Outliers were found only on the high side and all are plausible. I kept them. They do not drive attrition (8% and 11% against 16.7% overall). The attrition risk sits with lower-paid, junior and overtime-heavy employees.